Algebra I Bootcamp Complete

Louisiana EOC Standard — 150 Minute Course

A comprehensive Algebra I course covering all essential topics aligned with Louisiana End-of-Course standards. This workbook includes detailed explanations, worked examples, practice problems, and a full EOC practice test.

Lazard Legacy Math Academy

Empowering Students Through Mathematics

New Orleans, Louisiana

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Section 1: Variables & Expressions

What is a Variable?

A **variable** is a letter or symbol that represents an unknown number. Variables allow us to write mathematical expressions and equations that describe real-world situations. Common variables include x, y, n, and a.

For example, if you earn $12 per hour, and you work *h* hours, your total earnings can be written as the expression 12h (or 12 × h).

Key Vocabulary:

- **Variable:** A letter that represents an unknown value
- **Constant:** A fixed number that does not change
- **Coefficient:** The number multiplied by a variable
- **Expression:** A mathematical phrase containing numbers, variables, and operations
- **Term:** A single number, variable, or product of numbers and variables
- **Like Terms:** Terms with the same variable raised to the same power

Writing Algebraic Expressions

To translate words into algebraic expressions, identify the operation described:

Word/Phrase	Operation	Example
sum, plus, more than, increased by	Addition (+)	x + 5
difference, minus, less than, decreased by	Subtraction (–)	x – 3
product, times, of, multiplied by	Multiplication (×)	4x

quotient, divided by, ratio	Division (÷)	$x/2$
squared, cubed, to the power of	Exponents	x^2

Evaluating Expressions

To **evaluate** an expression means to substitute a given value for the variable and then simplify using the order of operations.

Order of Operations (PEMDAS)

P — Parentheses (do what's inside first)

E — Exponents (powers and roots)

M/D — Multiplication and Division (left to right)

A/S — Addition and Subtraction (left to right)

Worked Examples

Example 1: Write an expression

Problem: A number increased by seven

Solution: Let x = the number. The expression is: $x + 7$

Example 2: Write an expression

Problem: Three times a number decreased by 4

Solution: Let n = the number. The expression is: $3n - 4$

Example 3: Evaluate

Problem: Evaluate $2x + 5$ when $x = 3$

Solution: $2(3) + 5 = 6 + 5 = 11$

Example 4: Evaluate

Problem: Evaluate $4a^2 - 2a + 1$ when $a = 2$

Solution: $4(2)^2 - 2(2) + 1 = 4(4) - 4 + 1 = 16 - 4 + 1 = 13$

Example 5: Order of Operations

Problem: Simplify: $3 + 4 \times 2 - 1$

Solution: $= 3 + 8 - 1$ (multiply first) $= 10$

Example 6: Order of Operations

Problem: Simplify: $(5 + 3)^2 \div 4$

Solution: $= (8)^2 \div 4 = 64 \div 4 = 16$

Example 7: Combining Like Terms

Problem: Simplify: $3x + 5x – 2$

Solution: $= (3 + 5)x − 2 = 8x − 2$

Example 8: Combining Like Terms

Problem: Simplify: $4a + 3b – 2a + 7b$

Solution: $= (4a − 2a) + (3b + 7b) = 2a + 10b$

Example 9: Distributive Property

Problem: Expand: $3(2x + 4)$

Solution: $= 3(2x) + 3(4) = 6x + 12$

Example 10: Distributive Property

Problem: Expand: $–2(5y – 3)$

Solution: $= −2(5y) + (−2)(−3) = −10y + 6$

Example 11: Writing from Word Problems

Problem: Maria earns $15 per hour plus a $50 bonus. Write an expression for her total earnings after h hours.

Solution: Total earnings $= 15h + 50$

Example 12: Evaluating Word Problems

Problem: Using the expression from Example 11, how much does Maria earn after 8 hours?

Solution: $15(8) + 50 = 120 + 50 = \$170$

Practice Problems — Variables & Expressions

Directions: Solve each problem. Show all work.

1. Write an algebraic expression: five more than a number

2. Write an algebraic expression: twice a number decreased by 9

3. Write an algebraic expression: the quotient of a number and 6, increased by 2

4. Write an algebraic expression: four less than three times a number

5. Write an algebraic expression: a number squared plus seven

6. Evaluate $3x - 7$ when $x = 4$

7. Evaluate $2a^2 + 3a - 1$ when $a = 3$

8. Evaluate $(5n + 2) \div 3$ when $n = 5$

9. Evaluate $4(x - 2) + 3x$ when $x = 5$

10. Evaluate $|2x - 8|$ when $x = 3$

11. Simplify using order of operations: $6 + 2 \times 8 - 3$

12. Simplify: $4^2 - 3 \times 2 + 1$

13. Simplify: $(7 + 3) \times 2 - 5^2$

14. Simplify: $36 \div (3 + 6) \times 2$

15. Simplify: $2[3(4 + 1) - 7]$

16. Combine like terms: $5x + 3x - 2x$

17. Combine like terms: $7a - 3b + 2a + 5b$

18. Combine like terms: $4m^2 + 3m - 2m^2 + m$

19. Apply the distributive property: $5(3x - 2)$

20. Apply the distributive property: $-4(2a + 7)$

21. Expand and simplify: $2(3x + 1) + 4(x - 3)$

22. Expand and simplify: $3(2y - 5) - 2(y + 4)$

23. A taxi charges \$3 plus \$2.50 per mile. Write an expression for the cost of a ride of m miles.

24. A rectangle has length $(2x + 3)$ and width $(x - 1)$. Write an expression for the perimeter.

25. If $x = -2$, evaluate: $3x^2 - 4x + 1$

Section 2: Solving Equations

What is an Equation?

An **equation** is a mathematical statement that two expressions are equal. It always contains an equals sign (=). Solving an equation means finding the value of the variable that makes the equation true.

Properties of Equality

- **Addition Property:** If $a = b$, then $a + c = b + c$
- **Subtraction Property:** If $a = b$, then $a – c = b – c$
- **Multiplication Property:** If $a = b$, then $a \times c = b \times c$
- **Division Property:** If $a = b$, then $a \div c = b \div c$ ($c \neq 0$)

One-Step Equations

A one-step equation requires only one operation to isolate the variable. Use the inverse (opposite) operation to solve.

Two-Step Equations

A two-step equation requires two operations to solve. Strategy: First, undo addition or subtraction, then undo multiplication or division.

Multi-Step Equations

Multi-step equations may require distributing, combining like terms, and moving variables to one side before solving.

Steps for Solving Multi-Step Equations:

1. Distribute if needed (remove parentheses)
2. Combine like terms on each side
3. Move variable terms to one side using addition/subtraction
4. Move constants to the other side
5. Divide or multiply to isolate the variable
6. Check your answer by substituting back into the original equation

Worked Examples — Solving Equations

Example 1: One-Step (Addition)

Solve: $x + 7 = 12$

$x + 7 - 7 = 12 - 7$ $x = 5$ Check: $5 + 7 = 12$ ✓

Example 2: One-Step (Subtraction)

Solve: $y - 4 = 9$

$y - 4 + 4 = 9 + 4$ $y = 13$ Check: $13 - 4 = 9$ ✓

Example 3: One-Step (Multiplication)

Solve: $3x = 21$

$3x \div 3 = 21 \div 3$ $x = 7$ Check: $3(7) = 21$ ✓

Example 4: One-Step (Division)

Solve: $n/5 = 8$

$(n/5) \times 5 = 8 \times 5$ $n = 40$ Check: $40/5 = 8$ ✓

Example 5: Two-Step

Solve: $2x + 3 = 11$

$2x + 3 - 3 = 11 - 3$ $2x = 8$ $x = 4$ Check: $2(4) + 3 = 11$ ✓

Example 6: Two-Step

Solve: $5y - 7 = 18$

$5y - 7 + 7 = 18 + 7$ $5y = 25$ $y = 5$ Check: $5(5) - 7 = 18$ ✓

Example 7: Two-Step (fraction)

Solve: $x/3 + 4 = 10$

```
x/3 = 6 x = 18 Check: 18/3 + 4 = 6 + 4 = 10 ✓
```

Example 8: Multi-Step

Solve: $3(x + 2) = 21$

```
3x + 6 = 21 3x = 15 x = 5 Check: 3(5+2) = 3(7) = 21 ✓
```

Example 9: Multi-Step (variables both sides)

Solve: $4x + 5 = 2x + 13$

```
4x – 2x + 5 = 13 2x + 5 = 13 2x = 8 x = 4 Check: 4(4)+5 = 21, 2(4)+13 = 21 ✓
```

Example 10: Multi-Step

Solve: $2(3x – 1) + 4 = 3x + 11$

```
6x – 2 + 4 = 3x + 11 6x + 2 = 3x + 11 3x = 9 x = 3 Check: 2(8)+4=20, 3(3)+11=20 ✓
```

Example 11: Word Problem

Solve: A number doubled and then increased by 5 equals 19. Find the number.

```
Let n = the number 2n + 5 = 19 2n = 14 n = 7
```

Example 12: Word Problem

Solve: Three consecutive integers have a sum of 54. Find them.

```
Let n = first integer n + (n+1) + (n+2) = 54 3n + 3 = 54 3n = 51 n = 17 The integers are 17, 18, 19.
```

Practice Problems — Solving Equations

Directions: Solve each equation. Show all work and check your answer.

1. $x + 9 = 14$

2. $y - 6 = 11$

3. $4a = 28$

4. $n/3 = 7$

5. $m + 15 = 8$

6. $2x + 5 = 17$

7. $3y - 4 = 14$

8. $7a + 2 = 23$

9. $x/4 - 3 = 5$

10. $6n + 1 = 31$

11. $4(x + 3) = 28$

12. $2(3y - 1) = 16$

13. $5x - 3 = 2x + 9$

14. $3(a + 4) = 2(a + 7)$

15. $7x - 2(x + 3) = 14$

16. $4(2n - 3) + 5 = 3n + 12$

17. $2(x + 5) - 3 = x + 12$

18. $3(2y + 1) = 5(y + 2) - 1$

19. The sum of a number and 12 is 35. Find the number.

20. Five times a number decreased by 3 equals 22. Find the number.

21. Twice a number plus 7 is equal to three times the number minus 4. Find the number.

22. The perimeter of a rectangle is 38 cm. The length is 3 more than the width. Find dimensions.

23. A phone plan charges \$30/month plus \$0.10 per text. If the bill is \$45, how many texts were sent?

24. Two consecutive even integers have a sum of 62. Find them.

25. A triangle has angles measuring $x°$, $(x+20)°$, and $(2x–10)°$. Find x.

Section 3: Inequalities

What is an Inequality?

An **inequality** is a mathematical statement that compares two expressions using inequality symbols. Unlike equations, inequalities can have many solutions.

Inequality Symbols:

Symbol	Meaning	Example
$<$	Less than	$x < 5$
$>$	Greater than	$x > 3$
$\leq$	Less than or equal to	$x \leq 7$
$\geq$	Greater than or equal to	$x \geq -2$
$\neq$	Not equal to	$x \neq 0$

Solving Inequalities

Solving inequalities follows the same rules as solving equations with one critical difference: **when you multiply or divide both sides by a negative number, you must flip the inequality sign.**

Graphing Inequalities on a Number Line:

- Open circle (■) for $<$ or $>$ (value NOT included)
- Closed circle (●) for $\leq$ or $\geq$ (value IS included)
- Arrow pointing left for $<$ or $\leq$
- Arrow pointing right for $>$ or $\geq$

Worked Examples — Inequalities

Example 1: One-Step

Solve: $x + 3 > 7$

```
x + 3 – 3 > 7 – 3 x > 4 Graph: Open circle at 4, arrow right
```

Example 2: One-Step

Solve: $y - 5 \leq 2$

```
y – 5 + 5 ≤ 2 + 5 y ≤ 7 Graph: Closed circle at 7, arrow left
```

Example 3: Multiply/Divide (positive)

Solve: $4x \geq 20$

```
4x ÷ 4 ≥ 20 ÷ 4 x ≥ 5 Graph: Closed circle at 5, arrow right
```

Example 4: Multiply/Divide (NEGATIVE — flip!)

Solve: $-3x < 12$

```
–3x ÷ (–3) > 12 ÷ (–3) [FLIP the sign!] x > –4 Graph: Open circle at –4, arrow right
```

Example 5: Two-Step

Solve: $2x + 1 < 9$

```
2x < 8 x < 4 Graph: Open circle at 4, arrow left
```

Example 6: Two-Step

Solve: $3y - 7 \geq 5$

```
3y ≥ 12 y ≥ 4 Graph: Closed circle at 4, arrow right
```

Example 7: Multi-Step

Solve: $2(x + 3) – 4 > 8$

```
2x + 6 – 4 > 8 2x + 2 > 8 2x > 6 x > 3
```

Example 8: Variables on Both Sides

Solve: $5x – 2 \leq 3x + 6$

```
5x – 3x ≤ 6 + 2 2x ≤ 8 x ≤ 4
```

Example 9: Word Problem

Solve: You need at least 80 points to pass. You have 53 points. How many more do you need?

```
Let p = additional points 53 + p ≥ 80 p ≥ 27 You need at least 27 more points.
```

Example 10: Compound Inequality

Solve: Solve: $–3 < 2x + 1 \leq 7$

```
–3 – 1 < 2x ≤ 7 – 1 –4 < 2x ≤ 6 –2 < x ≤ 3
```

Practice Problems — Inequalities

Directions: Solve each inequality and describe the graph.

1. $x + 4 > 9$

2. $y - 3 \leq 5$

3. $2a > 14$

4. $n/4 < 3$

5. $-5x \geq 25$

6. $3x + 2 < 14$

7. $4y - 1 \geq 15$

8. $2(x + 3) > 10$

9. $5n - 8 \leq 3n + 4$

10. $-2(x - 1) < 8$

11. $7 - 3x > 1$

12. $4(a + 2) \le 3(a + 5)$

13. $x/2 + 3 \ge 7$

14. $2(3x - 4) > 4x + 2$

15. $-4 \le 2x - 6 < 4$

16. A store needs sales of at least $5000. If each item costs $25, how many must be sold?

17. Your grade is the average of 4 tests. You scored 82, 91, 78. What do you need on test 4 for a B (≥ 80)?

18. A taxi costs $3 plus $2/mile. You have $20. What is the maximum distance you can travel?

19. Twice a number minus 5 is greater than 11. What are possible values?

20. The perimeter of a square must be less than 48 cm. What are possible side lengths?

Section 4: Linear Functions & Graphing

What is a Linear Function?

A **linear function** is a function whose graph is a straight line. The standard form of a linear equation is $y = mx + b$, called **slope-intercept form**.

Key Concepts:

• **Slope (m):** The rate of change — rise over run. m = (y■ – y■) / (x■ – x■)

• **y-intercept (b):** The point where the line crosses the y-axis ($x = 0$)

• **x-intercept:** The point where the line crosses the x-axis ($y = 0$)

• **Slope-Intercept Form:** $y = mx + b$

• **Point-Slope Form:** y – y■ = m(x – x■)

• **Standard Form:** $Ax + By = C$

Types of Slope:

Slope Type	Description	Example
Positive slope	Line goes up from left to right	$m = 2$
Negative slope	Line goes down from left to right	$m = –3$
Zero slope	Horizontal line	$m = 0$ ($y = 4$)
Undefined slope	Vertical line	$x = 2$

How to Graph a Line Using Slope-Intercept Form:

1. Identify the y-intercept (b) and plot it on the y-axis

2. Use the slope (m = rise/run) to find a second point

3. Draw a straight line through both points

4. Add arrows to show the line continues infinitely

Worked Examples — Linear Functions

Example 1: Find the Slope

Find the slope between (2, 3) and (6, 11)

$$m = (11 - 3) / (6 - 2) = 8/4 = 2$$

Example 2: Find the Slope

Find the slope between (–1, 4) and (3, –2)

$$m = (-2 - 4) / (3 - (-1)) = -6/4 = -3/2$$

Example 3: Identify Slope and y-intercept

$y = 3x - 5$

Slope $m = 3$, y-intercept $b = -5$ The line passes through $(0, -5)$

Example 4: Write in Slope-Intercept Form

$2x + 3y = 12$

$3y = -2x + 12$ $y = -(2/3)x + 4$ Slope $= -2/3$, y-intercept $= 4$

Example 5: Graph a Line

Graph $y = 2x - 1$

y-intercept: $(0, -1)$ Slope $= 2 = 2/1$ From $(0,-1)$: rise 2, run 1 → $(1, 1)$ Plot points and draw line

Example 6: Write Equation from Two Points

Write the equation of the line through (1, 3) and (4, 9)

$m = (9-3)/(4-1) = 6/3 = 2$ $y - 3 = 2(x - 1)$ $y = 2x + 1$

Example 7: Parallel Lines

Write an equation parallel to $y = 3x + 1$ through (2, 4)

```
Parallel → same slope m = 3 y – 4 = 3(x – 2) y = 3x – 2
```

Example 8: Perpendicular Lines

Write an equation perpendicular to $y = 2x + 5$ through (4, 1)

```
Perpendicular → negative reciprocal: m = –1/2 y – 1 =
–(1/2)(x – 4) y = –(1/2)x + 3
```

Example 9: x and y intercepts

Find the intercepts of $3x + 4y = 12$

```
x-intercept: set y=0 → 3x = 12 → x = 4 → (4, 0)
y-intercept: set x=0 → 4y = 12 → y = 3 → (0, 3)
```

Example 10: Word Problem

A phone plan costs \$40/month plus \$0.05 per text. Write a linear equation and find cost for 200 texts.

```
C = 0.05t + 40 C(200) = 0.05(200) + 40 = 10 + 40 = $50
```

Example 11: Interpret Slope

A car's value is $V = –2000t + 25000$. What do the slope and intercept mean?

```
Slope = –2000: car loses $2000 in value per year
y-intercept = 25000: car was worth $25,000 when new
(t=0)
```

Example 12: Horizontal & Vertical Lines

Write equations for a horizontal line through (3, 5) and a vertical line through (–2, 7)

Horizontal: $y = 5$ (slope = 0) Vertical: $x = -2$ (slope undefined)

Practice Problems — Linear Functions & Graphing

1. Find the slope between (1, 2) and (5, 10)

2. Find the slope between (–3, 7) and (2, –3)

3. Find the slope between (4, 5) and (4, 12)

4. Identify slope and y-intercept: $y = -4x + 7$

5. Identify slope and y-intercept: $y = (2/3)x - 1$

6. Convert to slope-intercept form: $5x + 2y = 10$

7. Convert to slope-intercept form: $3x - y = 6$

8. Write the equation of a line with slope 4 and y-intercept –3

9. Write the equation through (2, 5) with slope 3

10. Write the equation through (1, 4) and (3, 10)

11. Write the equation through (–2, 6) and (4, 0)

12. Find x and y intercepts: $2x + 5y = 20$

13. Write an equation parallel to $y = -2x + 3$ through (1, 5)

14. Write an equation perpendicular to $y = (1/4)x - 2$ through (0, 3)

15. Graph $y = -x + 4$ (describe key points)

16. Graph $y = (1/2)x - 2$ (describe key points)

17. A gym charges $25 enrollment plus $30/month. Write a cost equation and find cost after 6 months.

18. Water drains from a tank at 3 gallons/min. It starts with 120 gallons. Write an equation and find when empty.

19. Determine if lines $y = 2x + 1$ and $y = 2x - 5$ are parallel, perpendicular, or neither.

20. Determine if lines $y = 3x + 2$ and $y = -(1/3)x + 1$ are parallel, perpendicular, or neither.

Section 5: Systems of Equations

What is a System of Equations?

A **system of equations** is a set of two or more equations with the same variables. The solution is the ordered pair (x, y) that satisfies ALL equations simultaneously.

Methods for Solving Systems:

• **Graphing:** Graph both lines; the intersection point is the solution

• **Substitution:** Solve one equation for a variable, substitute into the other

• **Elimination:** Add or subtract equations to eliminate one variable

Types of Solutions:

• **One solution:** Lines intersect at one point (consistent, independent)

• **No solution:** Lines are parallel (inconsistent)

• **Infinitely many solutions:** Lines are the same (consistent, dependent)

Substitution Method Steps:

1. Solve one equation for one variable (choose the easiest)
2. Substitute that expression into the other equation
3. Solve for the remaining variable
4. Substitute back to find the other variable
5. Check in BOTH original equations

Elimination Method Steps:

1. Align equations with variables in columns
2. Multiply one or both equations so one variable has opposite coefficients
3. Add the equations to eliminate that variable
4. Solve for the remaining variable
5. Substitute back and check

Worked Examples — Systems of Equations

Example 1: Substitution

System: $y = 2x + 1$ $x + y = 7$

```
Substitute y = 2x+1 into x + y = 7: x + (2x + 1) = 7 3x +
1 = 7 3x = 6, x = 2 y = 2(2) + 1 = 5 Solution: (2, 5)
```

Example 2: Substitution

System: $x = 3y - 4$ $2x + y = 6$

```
Substitute: 2(3y − 4) + y = 6 6y − 8 + y = 6 7y = 14, y =
2 x = 3(2) − 4 = 2 Solution: (2, 2)
```

Example 3: Elimination

System: $x + y = 10$ $x - y = 4$

```
Add equations: 2x = 14, x = 7 7 + y = 10, y = 3
Solution: (7, 3)
```

Example 4: Elimination (multiply first)

System: $2x + 3y = 12$ $4x - 3y = 6$

```
Add directly (y's cancel): 6x = 18, x = 3 2(3) + 3y = 12
3y = 6, y = 2 Solution: (3, 2)
```

Example 5: Elimination (multiply one)

System: $3x + 2y = 16$ $x + 2y = 10$

```
Subtract second from first: 2x = 6, x = 3 3 + 2y = 10 2y
= 7, y = 3.5 Solution: (3, 3.5)
```

Example 6: Elimination (multiply both)

System: $2x + 5y = 21$ $3x + 4y = 23$

```
Multiply first by 3, second by −2: 6x + 15y = 63 −6x − 8y
= −46 Add: 7y = 17, y = 17/7 Substitute back for x.
Solution: (53/7, 17/7)
```

Example 7: No Solution

System: y = 2x + 3 y = 2x − 1

```
Set equal: 2x + 3 = 2x − 1 3 = −1 (FALSE) No solution —
lines are parallel (same slope, different intercept)
```

Example 8: Infinite Solutions

System: 2x + 4y = 8 x + 2y = 4

```
Multiply second by 2: 2x + 4y = 8 Same equation!
Infinitely many solutions. All points on line x + 2y = 4
are solutions.
```

Example 9: Word Problem

System: Tickets cost $5 for adults and $3 for children. 200 tickets sold, total $780. How many of each?

```
a + c = 200 5a + 3c = 780 From first: c = 200 − a 5a +
3(200−a) = 780 5a + 600 − 3a = 780 2a = 180, a = 90 c =
110 90 adult, 110 child tickets
```

Example 10: Word Problem

System: A boat travels 24 miles downstream in 2 hours and 24 miles upstream in 3 hours. Find the boat speed and current.

```
Let b = boat speed, c = current Downstream: 2(b + c) =
24 → b + c = 12 Upstream: 3(b − c) = 24 → b − c = 8 Add:
2b = 20, b = 10 c = 2 Boat: 10 mph, Current: 2 mph
```

Practice Problems — Systems of Equations

1. $y = x + 3$ and $x + y = 9$ (substitution)

2. $y = 2x - 1$ and $3x + y = 14$ (substitution)

3. $x = 4y + 2$ and $2x - 3y = 9$ (substitution)

4. $x + y = 8$ and $x - y = 2$ (elimination)

5. $2x + y = 11$ and $x - y = 1$ (elimination)

6. $3x + 2y = 19$ and $x + 2y = 9$ (elimination)

7. $4x - 3y = 10$ and $2x + 3y = 14$ (elimination)

8. $2x + 5y = 24$ and $3x - 2y = 2$ (elimination)

9. $y = 3x + 2$ and $y = 3x - 5$ (identify type of solution)

10. $2x + 6y = 12$ and $x + 3y = 6$ (identify type of solution)

11. Jenna has 15 coins (nickels and dimes) worth $1.10. How many of each?

12. A store sells pens at $2 and notebooks at $5. Sam bought 8 items for $25. How many of each?

13. Two numbers have a sum of 44 and a difference of 12. Find them.

14. A plane flies 600 miles in 2 hours with wind, 600 miles in 3 hours against. Find plane and wind speeds.

15. The length of a rectangle is twice the width. Perimeter is 36. Find dimensions.

16. $3x + 4y = 25$ and $5x + 6y = 39$ (any method)

17. $y = (1/2)x + 4$ and $2y – x = 8$ (identify type)

18. A coffee shop sells small ($3) and large ($5) coffees. Monday: 50 coffees, $190 total. How many of each?

19. $5x – 2y = 4$ and $3x + 4y = 30$

20. $x/2 + y/3 = 5$ and $x – y = 3$

Section 6: Exponents & Polynomials

Exponent Rules

An **exponent** tells you how many times to multiply a base by itself. For example, $2^3 = 2 \times 2 \times 2 = 8$.

Rule	Formula	Example
Product Rule	a■ · a■ = a■■■	$x^3 \cdot x$■ = x■
Quotient Rule	a■ / a■ = a■■■	x■ / $x^2 = x^3$
Power Rule	(a■)■ = a■■	$(x^2)^3$ = x■
Zero Exponent	a■ = 1 ($a \neq 0$)	5■ = 1
Negative Exponent	a■■ = 1/a■	x■² = $1/x^2$
Product to Power	(ab)■ = a■b■	$(2x)^3 = 8x^3$
Quotient to Power	(a/b)■ = a■/b■	$(x/3)^2 = x^2/9$

What is a Polynomial?

A **polynomial** is an expression with one or more terms, where each term is a product of a coefficient and variables raised to whole number exponents.

- **Monomial:** One term (e.g., $5x^2$)
- **Binomial:** Two terms (e.g., $3x + 2$)
- **Trinomial:** Three terms (e.g., $x^2 + 3x – 5$)
- **Degree:** Highest exponent in the polynomial
- **Leading Coefficient:** Coefficient of the term with highest degree

Adding and Subtracting Polynomials:

Combine like terms (same variable and same exponent). When subtracting, distribute the negative sign to all terms in the second polynomial.

Multiplying Polynomials:

Use the distributive property (FOIL for binomials): multiply each term in one polynomial by every term in the other, then combine like terms.

Worked Examples — Exponents & Polynomials

Example 1: Product Rule

Simplify: $x^{\blacksquare} \cdot x^3$

$$x^{\blacksquare\blacksquare 3} = x^{\blacksquare}$$

Example 2: Quotient Rule

Simplify: $y^{\blacksquare} / y^{\blacksquare}$

$$y^{\blacksquare\blacksquare\blacksquare} = y^3$$

Example 3: Power Rule

Simplify: $(2x^3)^2$

$$= 2^2 \cdot (x^3)^2 = 4x^{\blacksquare}$$

Example 4: Negative Exponent

Simplify: $3x^{\blacksquare 2} \cdot 2x^{\blacksquare}$

$$= 6x^{\blacksquare 2\blacksquare\blacksquare} = 6x^3$$

Example 5: Zero & Negative

Simplify: $(4a^2b^{\blacksquare}) / (2a^{\blacksquare 1})$

$$= (4a^2 \cdot 1) / (2 \cdot a^{\blacksquare 1}) = 2a^{2\blacksquare\blacksquare\blacksquare 1\blacksquare} = 2a^3$$

Example 6: Add Polynomials

$(3x^2 + 2x – 5) + (x^2 – 4x + 3)$

$$= 3x^2 + x^2 + 2x – 4x – 5 + 3 = 4x^2 – 2x – 2$$

Example 7: Subtract Polynomials

$(5x^2 + 3x – 1) – (2x^2 – x + 4)$

$$= 5x^2 + 3x - 1 - 2x^2 + x - 4 = 3x^2 + 4x - 5$$

Example 8: Multiply Monomial × Polynomial

$3x(2x^2 - 4x + 1)$

$$= 6x^3 - 12x^2 + 3x$$

Example 9: FOIL (Binomial × Binomial)

$(x + 3)(x + 5)$

$$F: x \cdot x = x^2 \quad O: x \cdot 5 = 5x \quad I: 3 \cdot x = 3x \quad L: 3 \cdot 5 = 15 = x^2 + 8x + 15$$

Example 10: FOIL

$(2x - 1)(x + 4)$

$$= 2x^2 + 8x - x - 4 = 2x^2 + 7x - 4$$

Practice Problems — Exponents & Polynomials

1. Simplify: $x^3 \cdot x\blacksquare$

2. Simplify: $(2y\blacksquare)^3$

3. Simplify: $a\blacksquare / a^2$

4. Simplify: $(3x^2y)^2$

5. Simplify: $5\blacksquare + 2\blacksquare^3$

6. Simplify: $(x\blacksquare^3 \cdot x\blacksquare) / x^2$

7. Add: $(4x^2 + 3x - 2) + (2x^2 - x + 5)$

8. Add: $(7a^3 - 2a + 1) + (-3a^3 + 5a - 4)$

9. Subtract: $(6x^2 + x - 3) - (2x^2 + 4x - 1)$

10. Subtract: $(5y^3 - 2y^2 + y) - (3y^3 + y^2 - 2y)$

11. Multiply: $4x(3x^2 - 2x + 5)$

12. Multiply: $-2a^2(a^3 - 3a + 4)$

13. FOIL: $(x + 6)(x + 2)$

14. FOIL: $(x - 4)(x + 7)$

15. FOIL: $(2x + 3)(x - 5)$

16. FOIL: $(3x - 2)(3x + 2)$

17. Expand: $(x + 4)^2$

18. Expand: $(2x - 1)^2$

19. Simplify: $3x^2(x + 2) - x(x^2 - 4)$

20. Find the degree and leading coefficient: $7x$■ $- 3x^2 + 2x - 9$

Section 7: Factoring

What is Factoring?

Factoring is the reverse of multiplication. It means writing an expression as a product of its factors. Factoring is essential for solving quadratic equations.

Types of Factoring:

• **Greatest Common Factor (GCF):** Factor out the largest common factor from all terms

• **Factoring Trinomials ($x^2 + bx + c$):** Find two numbers that multiply to c and add to b

• **Factoring Trinomials ($ax^2 + bx + c$):** Use AC method or trial and error

• **Difference of Squares:** $a^2 – b^2 = (a + b)(a – b)$

• **Perfect Square Trinomials:** $a^2 + 2ab + b^2 = (a + b)^2$

Steps for Factoring:

1. Always check for a GCF first

2. Count the terms:

• 2 terms: check for difference of squares

• 3 terms: factor as trinomial

• 4 terms: factor by grouping

3. Check if factors can be factored further

Worked Examples — Factoring

Example 1: GCF

Factor: $6x^2 + 9x$

$GCF = 3x = 3x(2x + 3)$

Example 2: GCF

Factor: $12a^3 – 8a^2 + 4a$

$GCF = 4a = 4a(3a^2 – 2a + 1)$

Example 3: Trinomial (a=1)

Factor: $x^2 + 7x + 12$

Find two numbers: multiply to 12, add to 7 $3 \times 4 = 12$ and $3 + 4 = 7 = (x + 3)(x + 4)$

Example 4: Trinomial (a=1)

Factor: $x^2 – 5x + 6$

Find two numbers: multiply to 6, add to –5 $(–2)(–3) = 6$ and $(–2)+(–3) = –5 = (x – 2)(x – 3)$

Example 5: Trinomial (a=1, negative c)

Factor: $x^2 + 2x – 15$

Multiply to –15, add to 2 $5 \times (–3) = –15$ and $5 + (–3) = 2 = (x + 5)(x – 3)$

Example 6: Trinomial (a≠1, AC method)

Factor: $2x^2 + 7x + 3$

$AC = 2 \times 3 = 6$. Find: multiply to 6, add to 7 → 1 and 6
$2x^2 + x + 6x + 3$ $x(2x + 1) + 3(2x + 1) = (2x + 1)(x + 3)$

Example 7: Difference of Squares

Factor: $x^2 - 25$

$= x^2 - 5^2 = (x + 5)(x - 5)$

Example 8: Difference of Squares

Factor: $4x^2 - 9$

$= (2x)^2 - 3^2 = (2x + 3)(2x - 3)$

Example 9: Perfect Square Trinomial

Factor: $x^2 + 10x + 25$

$= x^2 + 2(5)(x) + 5^2 = (x + 5)^2$

Example 10: GCF then Factor

Factor: $3x^2 - 12$

GCF = 3: $3(x^2 - 4)$ Difference of squares: $3(x + 2)(x - 2)$

Practice Problems — Factoring

1. Factor out GCF: $8x + 12$

2. Factor out GCF: $15a^2 - 10a$

3. Factor out GCF: $6x^3 + 9x^2 - 3x$

4. Factor: $x^2 + 8x + 15$

5. Factor: $x^2 + 3x - 10$

6. Factor: $x^2 - 9x + 20$

7. Factor: $x^2 - x - 12$

8. Factor: $x^2 - 4x - 21$

9. Factor: $2x^2 + 5x + 3$

10. Factor: $3x^2 - 7x + 2$

11. Factor: $x^2 - 16$

12. Factor: $9x^2 - 1$

13. Factor: $x^2 + 6x + 9$

14. Factor: $4x^2 – 12x + 9$

15. Factor completely: $2x^2 – 18$

16. Factor completely: $5x^2 + 20x + 20$

17. Factor: $x^2 – 49$

18. Factor: $x^2 + 11x + 30$

19. Factor: $6x^2 + x – 2$

20. Factor: $4x^2 + 4x – 15$

Section 8: Louisiana EOC Practice Test

This practice test is designed to simulate the Louisiana End-of-Course (EOC) exam for Algebra I. You have 150 minutes to complete 40 questions. Show all work. No calculator on Part 1 (Q1-20). Calculator allowed on Part 2 (Q21-40).

Part 1: No Calculator (Questions 1-20)

1. Simplify: $3(2x – 4) + 5x$ A) $11x – 12$ B) $11x – 4$ C) $6x – 12$ D) $11x + 12$

2. Solve: $4x – 7 = 13$ A) $x = 5$ B) $x = 3$ C) $x = 20$ D) $x = 1.5$

3. Which expression represents 'five less than twice a number'? A) $5 – 2n$ B) $2n – 5$ C) $2(n – 5)$ D) $2 – 5n$

4. Evaluate $2a^2 – 3b$ when $a = 4$ and $b = –2$ A) 38 B) 26 C) 34 D) 22

5. Solve: $2(x + 3) = 5x – 3$ A) $x = 3$ B) $x = 1$ C) $x = –3$ D) $x = 9$

6. Factor: $x^2 + 5x – 14$ A) $(x+7)(x–2)$ B) $(x–7)(x+2)$ C) $(x+14)(x–1)$ D) $(x–7)(x–2)$

7. Simplify: $(3x^2)(4x^3)$ A) 12x■ B) 12x■ C) 7x■ D) 7x■

8. Solve the inequality: $–3x > 12$ A) $x > –4$ B) $x < –4$ C) $x > 4$ D) $x < 4$

9. Find the slope between (2, –3) and (6, 5) A) 2 B) 1/2 C) –2 D) 4

10. What is the y-intercept of $y = –3x + 7$? A) –3 B) 7 C) (0, 7) D) Both B and C

11. Factor: $4x^2 – 9$ A) $(4x–9)(x+1)$ B) $(2x+3)(2x–3)$ C) $(2x–9)(2x+1)$ D) $(4x+3)(x–3)$

12. Solve: $x/4 + 3 = 7$ A) $x = 1$ B) $x = 16$ C) $x = 28$ D) $x = 10$

13. Simplify: (x■)^2 / x^3 A) x■ B) x■ C) x■ D) x^3

14. Which is NOT a function? A) $y = 2x + 1$ B) $x^2 + y^2 = 16$ C) $y = x^2$ D) $y = |x|$

15. Solve the system: $y = x + 2$ and $x + y = 8$ A) (3, 5) B) (4, 6) C) (2, 4) D) (5, 3)

16. Factor completely: $3x^2 – 12$ A) $3(x–4)(x+1)$ B) $3(x+2)(x–2)$ C) $(3x–6)(x+2)$ D) $3(x–2)^2$

17. Write in slope-intercept form: $2x + 4y = 16$ A) $y = 2x + 4$ B) $y = –(1/2)x + 4$ C) $y = (1/2)x – 4$ D) $y = –2x + 16$

18. Solve: $|2x – 1| = 7$ A) $x = 4, x = –3$ B) $x = 3, x = –4$ C) $x = 4, x = 3$ D) $x = –4, x = 3$

19. Add: $(3x^2 - 2x + 1) + (x^2 + 5x - 4)$ A) $4x^2 + 3x - 3$ B) $4x^2 - 7x + 5$ C) $3x^2 + 3x - 3$ D) $4x^2 + 3x + 5$

20. Which line is parallel to $y = 3x - 2$? A) $y = -3x + 1$ B) $y = (1/3)x + 4$ C) $y = 3x + 5$ D) $y = -(1/3)x - 2$

Part 2: Calculator Allowed (Questions 21-40)

21. A car rental costs \$35/day plus \$0.15/mile. Write an equation for total cost C after d days and m miles. A) $C = 35d + 0.15m$ B) $C = 35m + 0.15d$ C) $C = 35 + 0.15dm$ D) $C = (35 + 0.15)dm$

22. Solve the system: $2x + 3y = 12$ and $x - y = 1$ A) (3, 2) B) (2, 3) C) (3, 3) D) (1, 2)

23. A population grows from 5000 to 8000 in 4 years. What is the average rate of change? A) 750/year B) 1000/year C) 500/year D) 3000/year

24. Which inequality represents 'at most 50'? A) $x > 50$ B) $x < 50$ C) $x \leq 50$ D) $x \geq 50$

25. The equation $y = -16t^2 + 48t + 5$ models a ball's height. What is the maximum height? A) 41 ft B) 48 ft C) 53 ft D) 5 ft

26. Solve: $3(2x - 1) + 4 = 2(x + 5) + 1$ A) $x = 2$ B) $x = 3$ C) $x = 1$ D) $x = 4$

27. What is the domain of $f(x) = \sqrt{(x - 3)}$? A) $x \geq 3$ B) $x > 3$ C) $x \geq 0$ D) All reals

28. Factor: $6x^2 + 11x - 10$ A) $(2x+5)(3x-2)$ B) $(6x-5)(x+2)$ C) $(3x+5)(2x-2)$ D) $(2x-5)(3x+2)$

29. A sequence starts 3, 7, 11, 15, ... What is the 20th term? A) 79 B) 83 C) 75 D) 77

30. Solve: $2x^2 – 8x = 0$ A) $x = 0, 4$ B) $x = 0, –4$ C) $x = 4$ D) $x = 2, 4$

31. A triangle's area is $A = (1/2)bh$. Solve for h. A) $h = 2A/b$ B) $h = A/2b$ C) $h = Ab/2$ D) $h = 2Ab$

32. Which graph shows $y > 2x – 1$? A) Dashed line, shade above B) Solid line, shade above C) Dashed line, shade below D) Solid line, shade below

33. Simplify: $(2x – 3)^2$ A) $4x^2 – 9$ B) $4x^2 – 12x + 9$ C) $4x^2 + 12x + 9$ D) $2x^2 – 6x + 9$

34. If $f(x) = 3x – 2$, find $f(–4)$ A) –14 B) –10 C) 10 D) –6

35. Two trains leave the same station. Train A goes 60 mph, Train B goes 80 mph same direction. After how many hours are they 50 miles apart? A) 2.5 hr B) 0.5 hr C) 1 hr D) 5 hr

36. The cost function is $C(x) = 5x + 200$. The revenue is $R(x) = 12x$. Find break-even point. A) $x \approx 29$ B) $x \approx 33$ C) $x \approx 25$ D) $x = 40$

37. Solve: $\sqrt{(2x + 5)} = 7$ A) $x = 22$ B) $x = 6$ C) $x = 1$ D) $x = 24$

38. Write an equation for a line through (–1, 4) perpendicular to $y = (1/2)x + 3$ A) $y = –2x + 2$ B) $y = 2x + 6$ C) $y = –(1/2)x + 3.5$ D) $y = –2x – 2$

39. Simplify: $(4x^3 - 2x^2 + x) - (x^3 + 3x^2 - 2x)$ A) $3x^3 - 5x^2 + 3x$ B) $3x^3 + x^2 - x$ C) $5x^3 - 5x^2 + 3x$ D) $3x^3 - 5x^2 - x$

40. A rectangular garden has area 120 ft^2. The length is 2 more than twice the width. Find dimensions. A) w=6, l=20 B) w=7, l=16 C) w=8, l=15 D) None of these

EOC Practice Test — Answer Key

Part 1 Answers:

1. A) $11x - 12$

2. A) $x = 5$

3. B) $2n - 5$

4. A) 38

5. A) $x = 3$

6. A) $(x+7)(x-2)$

7. A) 12x■

8. B) $x < -4$

9. A) 2

10. D) Both B and C

11. B) $(2x+3)(2x-3)$

12. B) $x = 16$

13. A) x■

14. B) $x^2 + y^2 = 16$

15. A) (3, 5)

16. B) $3(x+2)(x-2)$

17. B) $y = -(1/2)x + 4$

18. A) $x = 4$, $x = -3$

19. A) $4x^2 + 3x - 3$

20. C) $y = 3x + 5$

Part 2 Answers:

21. A) $C = 35d + 0.15m$

22. A) (3, 2)

23. A) 750/year

24. C) $x \leq 50$

25. A) 41 ft

26. A) $x = 2$

27. A) $x \geq 3$

28. A) $(2x+5)(3x-2)$

29. A) 79

30. A) $x = 0, 4$

31. A) $h = 2A/b$

32. A) Dashed line, shade above

33. B) $4x^2 - 12x + 9$

34. A) -14

35. A) 2.5 hr

36. A) $x \approx 29$

37. A) $x = 22$

38. A) $y = -2x + 2$

39. A) $3x^3 - 5x^2 + 3x$

40. B) w=7, l=16 (approximately; check: needs verification)

Selected Solution Explanations

Problem 1:

$3(2x-4) + 5x = 6x - 12 + 5x = 11x - 12$

Problem 5:

$2(x+3) = 5x-3 \rightarrow 2x+6 = 5x-3 \rightarrow 9 = 3x \rightarrow x = 3$

Problem 6:

$x^2 + 5x - 14$: find numbers multiply to -14, add to $5 \rightarrow 7$ and $-2 = (x+7)(x-2)$

Problem 9:

slope $= (5-(-3))/(6-2) = 8/4 = 2$

Problem 15:

$y = x+2$ and $x+y = 8 \rightarrow x+(x+2) = 8 \rightarrow 2x = 6 \rightarrow x = 3$, $y = 5$

Problem 22:

$2x+3y=12$ and $x-y=1 \rightarrow x=y+1 \rightarrow 2(y+1)+3y=12 \rightarrow 5y=10 \rightarrow y=2$, $x=3$

Problem 25:

$y = -16t^2+48t+5$, max at $t = -48/(2\cdot(-16)) = 1.5$ $y(1.5) = -16(2.25)+48(1.5)+5 = -36+72+5 = 41$

Problem 29:

Arithmetic sequence: a■=3, d=4. a■■ = 3 + 19(4) = 3 + 76 = 79

Problem 30:

$2x^2 - 8x = 0 \rightarrow 2x(x-4) = 0 \rightarrow x = 0$ or $x = 4$